which should contain, at a minimum, each value of the calculation parameters and the calculation process used to determine the attained EEDI; and

.7 calculated values of the attained EEDIweather and fw value (not equal to 1.0), if those values are calculated, based on the EEDI Calculation Guidelines.

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Electric power table should be validated separately, taking into account guidelines set out in

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if those values are calculated, based on the EEDI Calculation Guidelines.

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Electric power table should be validated separately, taking into account guidelines set out in appendix 2 to these Guidelines.

A sample of an EEDI Technical File is provided in the appendix 1 to these Guidelines.

4.2.3 If dual fuel engines are installed on the ship, the C F factor and the Specific Fuel Consumption of gas fuel should be used if gas fuel will be used as the primary fuel on the ship. In order to verify this, the following information should be provided:

.1 The use of boil-off gas or capacities of the gas fuel storage tanks, and the capacities of fuel oil storage tanks.

.2 Arrangements of bunkering facilities for gas fuel in the intended operational area of the ship.

4.2.4 The SFC of the main and auxiliary engines should be quoted from the approved NOx Technical File and should be corrected to the value corresponding to the ISO standard reference conditions using the standard lower calorific value of the fuel oil (42,700kJ/kg), referring to ISO 15550:2002 and ISO 3046-1:2002. For the confirmation of the SFC, a copy

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reference conditions using the standard lower calorific value of the fuel oil (42,700kJ/kg), referring to ISO 15550:2002 and ISO 3046-1:2002. For the confirmation of the SFC, a copy of the approved NOx Technical File and documented summary of the correction calculations should be submitted to the verifier. In case NO x Technical File has not been approved at the time of the application for initial survey, the test reports provided by manufacturers should be used. In this case, at the time of the sea trial verification, a copy of the approved NO x Technical File and documented summary of the correction calculations should be submitted to the verifier.

Note: SFC in the NO x Technical File are the values of a parent engine, and the use of such value of SFC for the EEDI calculation for member engines may have the following technical issues for further consideration:

.1 The definition of "member engines " given in NO x Technical File is broad and specification of engines belonging to the same group/family may vary; and

.2 The rate of NO x emission of the parent engine is the highest in the group/family

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SOURCE: MEPC.214(63).pdf
.1 The definition of "member engines " given in NO x Technical File is broad and specification of engines belonging to the same group/family may vary; and

.2 The rate of NO x emission of the parent engine is the highest in the group/family i.e. CO2 emission, which is in the trade-off relationship with NO x emission, can be lower than the other engines in the group/family.

4.2.5 For ships to which regulat ion 21 of MARPOL Annex VI applies, the power curves used for the preliminary verification at the design stage should be based on reliable results of tank test. A tank test for an individual ship may be omitted based on technical justifications

such as availability of the results of tank tests for ships of the same type. In addition, omission of tank tests is acceptable for a ship for which sea trials will be carried under the condition as specified in paragraph 2.2 of the EEDI Calculation Guidelines, upon agreement

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omission of tank tests is acceptable for a ship for which sea trials will be carried under the condition as specified in paragraph 2.2 of the EEDI Calculation Guidelines, upon agreement of the shipowner and shipbuilder and with approval of the verifier. For ensuring the quality of tank tests, ITTC quality system should be taken into account. Model tank test should be witnessed by the verifier.

Note: It would be desirable in the future that an organization conducting a tank test be authorized.

4.2.6 The verifier may request the submitter for additional information on top of those contained in Technical File, as necessary, to examine the calculation process of the attained EEDI. The estimation of the ship speed at the design stage much depends on each shipbuilder's experiences, and it may not be practicable for any person/organization other than the shipbuilder to fully examine the technical aspects of experience-based parameters such as the roughness coefficient and wake scaling coefficient. Therefore, the preliminary verification should focus on the calculation process of the attained EEDI to ensure that it is

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such as the roughness coefficient and wake scaling coefficient. Therefore, the preliminary

verification should focus on the calculation process of the attained EEDI to ensure that it is technically sound and reasonable and follows regulation 20 of MARPOL Annex VI and the EEDI Calculation Guidelines.

Note 1: A possible way forward for more robust verification is to establish a standard methodology of deriving the ship speed from the outcomes of tank test, by setting standard values for experience-based correction factors such as roughness coefficient and wake scaling coefficient. In this way, ship- by-ship performance comparison could be made more objectively by excluding the possibility of arbitrary setting of experience-based parameters. If such standardization is sought, this would have an implication on how the ship speed adjustment based on sea trial results should be conducted in accordance with paragraph 4.3.8 of these Guidelines.

Note 2: A joint industry standard to support the method and role of the verifier will be developed.

4.2.7 Additional information that the verifier may request the submitter to provide directly

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paragraph 4.3.8 of these Guidelines.

Note 2: A joint industry standard to support the method and role of the verifier will be developed.

4.2.7 Additional information that the verifier may request the submitter to provide directly
to it includes but not limited to:

.1 descriptions of a tank test facility; this should include the name of the facility, the particulars of tanks and towing equipment, and the records of calibration of each monitoring equipment;

.2 lines of a model ship and an actual ship for the verification of the appropriateness of the tank test; the lines (sheer plan, body plan and half-breadth plan) should be detailed enough to demonstrate the similarity between the model ship and the actual ship;

.3 lightweight of the ship and displacement table for the verification of the deadweight;

.4 detailed report on the method and results of the tank test; this should include at least the tank test results at sea trial condition and under the condition as specified in paragraph 2.2 of the EEDI Calculation Guidelines;

.5 detail ed calculation process of the ship speed, which should include the

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include at least the tank test results at sea trial condition and under the condition as specified in paragraph 2.2 of the EEDI Calculation Guidelines;

.5 detail ed calculation process of the ship speed, which should include the

estimation basis of experience-based parameters such as roughness coefficient, wake scaling coefficient; and

.6 reasons for exempting a tank test, if applicable; this should include lines and tank test results of the ships of same type, and the comparison of the

principal particulars of such ships and the ship in question. Appropriate technical justification should be provided explaining why the tank test is unnecessary.

4.2.8 The verifier should issue the report on the "Preliminary Verification of EEDI" aft er it has verified the attained EEDI at the design stage in accordance with paragraphs 4.1 and 4.2 of these Guidelines.

4.3 Final verification of the attained EEDI at sea trial